

Geralt Heart

Austin TX | geraldheart01@gmail.com | 956-802-1359 | github.com/GHeart01 | linkedin.com/in/geraldheart001 | geraldheart.com

EDUCATION

University of Colorado Boulder | GPA: 3.9/4.0

Expected May 2027

Master of Science in Computer Science

Boulder, CO

University of Texas Rio Grande Valley | GPA: 3.5/4.0

July 2017 – May 2020

Bachelor of Science in Applied Mathematics

Edinburg, TX

PROJECTS

Regression Testing Framework | *python, pytest, GitHub Actions*

<https://github.com/GHeart01/tile-gps-regression-testing>

- Designed 52 automated unit and regression tests covering 4 system modules (TILE, EMQuest GPS, Spectra Python, and Aurora systems) achieving 100% test validation with execution time under 0.1 seconds.
- Validated Android GPS module functionality including location accuracy, signal strength, and RTK/DGPS compatibility.
- Implemented Python 3.9+ integration tests for ancillary state and data persistence, data serialization, JSON validation, and configuration management
- Created CI/CD pipeline using GitHub Actions with automated testing, coverage analysis, and multi-version support

Pricing Web Scraper | *PostgreSQL, Python, BeautifulSoup, REST API*

<https://github.com/GHeart01/Web-Scraper-Clothing>

- Implemented a PostgreSQL relational database scheme for indexing the price of men's clothing across 4 different major retailers.
- Built JSON serialization pipeline for storing price history with timestamps.
- Utilized rate limiting to manage anti-scraping protections (403 errors) and evaluated performance against REST API alternatives.
- Created master scraper controller that aggregates results from multiple sources, calculates price comparisons, and exports normalized JSON data compatible with PostgreSQL

Customer Segmentation & Clustering Analysis | *NumPy, pandas, seaborn, matplotlib, TensorFlow*

github.com/GHeart01/Wholesale_Customer_Clustering

- Built a customer segmentation pipeline using K-Means and Hierarchical Clustering to identify purchasing behavior patterns.
- Evaluated clustering quality using silhouette scores between multiple cluster counts, provided heatmap/pair plot visualizations to demonstrate feature relationships.
- Applied hyperparameter tuning across several different models to evaluate and cross-validate performance.

WORK EXPERIENCE

Guest Teacher

Aug 2025 – Present

Kokua Education

Austin, TX

Mathematics Tutor

Sept 2018 – June 2024

South Texas College / Freelance

McAllen, TX

- Performed one-on-one and group instruction for mathematics and statistics courses including College Algebra, Calculus I, II, III, Elementary Statistics, Differential Equations, and other mathematics classes.

SKILLS

Programming Languages: Python, SQL, C++, JavaScript, MATLAB

Machine Learning: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, SciPy

Frameworks & Libraries: React, pytest, Three.js, GSAP

Databases: PostgreSQL, MySQL

Cloud & DevOps: AWS Lightsail, Docker, Kubernetes